

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

the de Broglie wavelength of a particle is given by the formula $\lambda = h / \sqrt{2 m E}$, where λ is the wavelength , h is the Planck's constant , m is the mass of the particle , and E is its kinetic energy .

first , we need to convert the kinetic energy from eV to Joules : $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$.

then , we can use the mass of an electron ($9.11 \times 10^{-31} \text{ kg}$) and the Planck's constant ($6.626 \times 10^{-34} \text{ J s}$) to calculate the de Broglie wavelength .

$\lambda = h / \sqrt{2 m E} = (6.626 \times 10^{-34} \text{ J s}) / \sqrt{2 * 9.11 \times 10^{-31} \text{ kg} * 1.6 \times 10^{-19} \text{ J}} = 1.22 \times 10^{-9} \text{ m}$.

converting this to cm , we get $\lambda = 1.22 \times 10^{-9} \text{ m} * 100 \text{ cm/m} = 1.22 \times 10^{-7} \text{ cm}$.

</think><answer> G . 1.23×10^{-7} cm